

SENTINEL-E: Anytime-Valid Sequential Detection for Streaming Illegal Dumping Surveillance

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Abstract

Computer-vision systems for illegal dumping are trained and evaluated as fixed-sample-size classifiers: given a short clip, emit one dumping / no-dumping decision and report clip-level precision and recall. Every entry in the 2026 IWDD contest on the Mivia-IWDD benchmark follows this protocol. Deployment does not look like that. A municipal camera runs for weeks, and the quantity an operator manages is the number of false alarms accumulated over the whole monitoring period, because each one costs an enforcement dispatch. Alerting the first time a threshold is crossed on a continuous stream is optional stopping, and it inflates the true false-alarm rate by an amount clip-level evaluation cannot show. We measure it and find it severe: a per-frame rule fed p-values that are individually valid at the 1% level raises at least one false alarm on 100% of 40-minute null streams. Our main contribution is an algorithm, not a combination of existing ones. Every sequential change detector in use — CUSUM, Shiryaev–Roberts, e-detectors, and the recent non-partitioned e-detectors — models a *persistent* switch from one law to another that lasts forever. A dumping event is an *episode*: it begins, flickers because the offender is intermittently occluded, ends, and recurs weeks later at the same site. We replace the mixture over changepoint locations with a mixture over episodes, realised as a two-state Markov chain whose forward recursion collapses a sum over 2^t trajectories into two numbers. The resulting *episodic e-process* is an exact non-negative supermartingale, so Ville’s inequality gives time-uniform false-alarm control with no new machinery; it costs $O(1)$ per frame, where a general mixture over active starts costs $O(t)$; and it contains the changepoint mixture exactly as the boundary case in which episodes never end, so it is a strict generalisation along an axis the change-detection literature has left fixed at infinity. The recursion also returns, for free, the posterior probability that a camera is inside an episode — a bounded, cross-camera-comparable quantity that becomes the message a camera graph exchanges. Around this core we build a per-camera conformal layer that is calibrated conditionally on observable context and corrected by exact Beta order-statistic levels, and a fleet layer in which a graph network steers each camera’s episode hazard through a *predictable* modulation — provably unable to damage the guarantee — with e-BH combining evidence under arbitrary dependence. On recurring episodes the episodic mixture cuts the missed-detection rate from 0.010 to 0.000 against the changepoint mixture; on a single persistent change the two agree, as a strict generalisation should. Across the baseline comparison SENTINEL-E is the only method that holds its nominal false-alarm level at all, and it costs 0.61 μ s per frame, 0.0016% of the detector it wraps. We also report where it stops holding: a shift in an unmodelled covariate beyond $\times 2.0$ inflates the rate unless the shift-weighting safeguard is engaged, which restores validity at a steep cost in power.

Keywords: illegal dumping detection; anytime-valid inference; e-values; conformal prediction; sequential change detection; false discovery rate; video surveillance; edge deployment

1 Introduction

Illegal dumping is a persistent municipal problem with an obvious computer-vision framing, and the field has adopted that framing consistently: collect clips, label them as containing a dumping

event or not, train a video classifier, report clip-level accuracy. Mivia-IWDD-500 [Greco et al., 2026] — the first public dataset for illegal waste dumping detection, with 250 positive and 250 negative surveillance videos and onset timestamps — gave the field a shared benchmark on these terms, and the 2026 IWDD contest built on it attracted a strong and varied field: X3D with temporal localisation [Almorsy and Torki, 2026], cross-fold ensembles of Hiera and VideoMAE [Kim et al., 2026], a three-headed classifier [Scarrica et al., 2026], a vision–language temporal analysis [Kairanbay, 2026], and a temporal-spatial novelty detector [Zbuce et al., 2026]. Every one of them reports per-video precision, recall and F_1 together with a timestamp error. Where “false alarms” appear, they are counted per clip — “six false alarms on thirty negative videos” [Almorsy and Torki, 2026] — which is specificity, not a rate over a monitoring period.

The framing does not match deployment. A camera watching a lay-by does not receive a curated ten-second clip and return a verdict; it produces frames continuously for weeks, and the system alerts the first time its evidence crosses a threshold. What the municipality manages is not per-clip precision but the number of enforcement dispatches wasted over a season. Those two quantities are separated by a step the clip-level protocol never exercises: *repeatedly* testing a stream and stopping at the first exceedance is optional stopping, and it invalidates the level at which any fixed threshold was calibrated.

The magnitude of the gap is easy to state and, we find, easy to underestimate. A rule that raises a false alarm on each frame with probability q raises at least one over T frames with probability $1 - (1 - q)^T$, which tends to one for any $q > 0$. Persistence filters and score smoothing change q ; they do not change the limit. Feeding a per-frame rule p-values that are individually and provably valid at the 1% level — so that nothing is wrong with the calibration — produces at least one false alarm on 100% of 40-minute null streams (Figure 1 (a)); a raw score threshold tuned to a 10^{-3} per-frame rate reaches 0.640 within 0.2 hours and 1.000 by 2.7 hours (Figure 1 (b)). A clip-level evaluation cannot see this, because it never runs the detector for two hours.

1.1 Why the existing sequential machinery is the wrong shape

The tools for optional stopping are mature. Ville’s inequality [Ville, 1939] bounds the probability that a non-negative supermartingale *ever* exceeds a level, and the game-theoretic literature [Shafer and Vovk, 2019, Ramdas et al., 2023, Grünwald et al., 2024] turns that into a construction: bet against the null, and the accumulated wealth is a time-uniform certificate. Conformal prediction [Vovk et al., 2005] supplies distribution-free p-values to bet on, e-detectors [Shin et al., 2024] adapt the machinery to change detection, non-partitioned e-detectors [Saha and Ramdas, 2026] sharpen it to composite pre-change classes, and e-BH [Wang and Ramdas, 2022] combines evidence across streams under arbitrary dependence. To our knowledge none of it has been applied to waste, dumping or pollution surveillance.

Importing it unchanged would not be enough, and the reason is structural rather than a matter of tuning. *Every one of these procedures models a persistent change.* At an unknown frame ν the law switches from P_0 to P_1 and stays there for the rest of time; the mixtures are over ν alone, and the optimality theory, thresholds and delay bounds all presume it.

An illegal dumping event is not a persistent change. It is an *episode*. A van pulls up, the offender unloads for forty seconds and drives away. Inside the episode the signal is *intermittent*: the offender is occluded by the vehicle, steps out of frame, returns. Then the episode ends, the stream reverts to its null behaviour, and — because a site that has been used once is used again — another episode follows weeks later.

Modelling that as a persistent change is not a small mismatch. A changepoint-mixture wealth process keeps betting at full aggression after the episode is over, handing back what the episode earned; and because it treats every frame after ν as drawn from the post-change law, the quiet frames *inside* the episode count against it rather than being expected. Both effects lengthen delay, and the second worsens as intermittency increases — which is the regime this problem

lives in. Empirically, over four recurring episodes per camera the changepoint mixture misses 0.010 of events where the episodic construction misses 0.000.

1.2 Contributions

- **The episodic e-process** (Section 4.2), our main contribution. Mixing over episodes rather than changepoints, via a two-state Markov chain, gives a wealth process that is (i) an exact non-negative supermartingale, so Ville’s inequality applies unchanged; (ii) computable in $O(1)$ time and memory per frame by an HMM forward recursion, where a general mixture over active starts costs $O(t)$ per observation [Saha and Ramdas, 2026]; and (iii) a strict generalisation, equal to the changepoint mixture exactly when the episode-end rate is zero (Proposition 4).
- **A quantity worth passing between cameras.** The same recursion returns the posterior probability that a camera is currently inside an episode. Unlike accumulated wealth it is bounded in $[0, 1]$ and comparable across cameras with different histories, which is what lets a message-passing layer aggregate it without one long-running camera swamping its neighbours (Section 4.3).
- **Exact calibration-conditional conformal levels** (Theorem 2). Each attainable p-value level is replaced by its exact Beta order-statistic upper confidence bound rather than inflated additively. At $n = 2000$, $\delta = 10^{-3}$ the smallest usable p-value becomes 0.0037 instead of the 0.051 that Dvoretzky–Kiefer–Wolfowitz permits — a $14\times$ improvement precisely where a betting detector earns its wealth, worth a factor of 8.3 in detection delay.
- **A learned spatial prior that provably cannot break the guarantee** (Theorem 5). Because the graph network’s output enters only as a predictable modulation of the episode hazard and the betting stake, the wealth remains a supermartingale whatever the network outputs — even badly trained or handed the wrong topology. It can move the delay and nothing else.
- **An evaluation that measures guarantees rather than asserting them**, over up to 1500 Monte-Carlo streams per operating point, against six baselines including a reimplementation of the closest generic prior art [Khan and Syed, 2026]; a component ablation in which 5 of 19 variants fail to deliver the level they claim; and a cost measurement putting the guarantee layer at 0.0016% of its backbone. We also correct a measurement pitfall that affects such comparisons generally: detection delay conditioned on detection is averaged over a different subset of runs for each method and systematically flatters the ones that miss the hard events, so we report a censored delay alongside it (Section 3).
- **A negative result we would have preferred not to have.** We introduced a dilated betting function to model within-episode intermittency, and the ablation shows the κ mixture already handles it: dilation costs delay and never repays it (Section 4.2.2). It is off by default and reported rather than buried.

What we do not claim. The changepoint mixture with retained tail mass that our construction reduces to is not new: it is the PFA-controlled e-detector of Saha and Ramdas [2026], which already delivers $\mathbb{P}(\tau < \infty) \leq \alpha$ for a mixture over start times with weights summing to one. Our contribution is the episode axis on top of it, and the $O(1)$ recursion that axis happens to make available. The conformal and e-BH layers are existing machinery, used as intended and credited as such; Section 7 states plainly what the evaluation does and does not establish.

2 Related work

Waste and dumping detection. Image-level litter detection is well served by TACO [Proença and Simões, 2020] and ZeroWaste [Bashkistrova et al., 2022]; UAV and wide-area monitoring by UAVVaste [Kraft et al., 2021] and AerialWaste [Torres and Fraternali, 2023]; marine pollution by MARIDA [Kikaki et al., 2022] and MADOS [Kikaki et al., 2024]. Video-based dumping detection now has a public benchmark and an active contest around it: on Mivia-IWDD-500 [Greco et al., 2026], the 2026 IWDD entries span 3D CNNs with temporal localisation [Almorsy and Torki, 2026], cross-fold transformer ensembles [Kim et al., 2026], multi-head classifiers [Scarrica et al., 2026], vision-language temporal reasoning [Kairanbay, 2026] and novelty detection [Zbuca et al., 2026]. These are strong backbones and our work does not compete with them: SENTINEL-E consumes a frozen backbone’s scores, so a better backbone and this work compound. What none of them reports — and what the clip-level protocol they share cannot express — is a false-alarm rate over a monitoring horizon, an average run length, or a detection delay under continuous monitoring.

Sequential change detection. CUSUM [Page, 1954], Shiryaev’s Bayesian statistic [Shiryaev, 1963] and Shiryaev–Roberts [Roberts, 1966] are the classical procedures, with optimality theory due to Lorden [1971] and Moustakides [1986]. Their guarantees are asymptotic average-run-length statements presuming a correctly specified pre-change law; detector scores from real footage are autocorrelated, heavy-tailed and diurnally non-stationary, so the presumption fails and both lose control in our experiments. Shin et al. [2024] give a nonparametric e-detector framework with non-asymptotic ARL bounds, and Saha and Ramdas [2026] extend it to composite pre-change classes, including a PFA-controlled variant that mixes over start times with weights summing to one and thereby achieves $\mathbb{P}(\tau < \infty) \leq \alpha$. That variant is exactly what our Proposition 4 reduces to, and we claim no credit for it.

The difference is the change model. All of these mix over the changepoint *location* of a persistent shift; none of them models an episode that ends, let alone one that recurs or that is internally intermittent. That is the axis Section 4.2 adds. A secondary but practical difference is cost. Evaluated exactly across all active starts, a non-partitioned e-detector is $O(t)$ per observation [Saha and Ramdas, 2026]; under a geometric prior over start times the sum telescopes to an $O(1)$ recursion, which is the special case of Equation (4) at $\eta = 0$. The point is that the episodic process retains that $O(1)$ cost *while adding the episode-end axis* — the extra generality is free arithmetically as well as statistically, which matters when the horizon is a month of video and the hardware is a camera.

Anytime-valid inference and e-values. Testing by betting [Shafer, 2021, Ramdas et al., 2023, Grünwald et al., 2024] turns Ville’s inequality into a design principle; Waudby-Smith and Ramdas [2024] develop the betting strategies we adapt. Wang and Ramdas [2022] introduce e-BH, which controls FDR under arbitrary dependence and underpins our fleet layer; Vovk and Wang [2021] characterise e-value merging. Closest to this work in spirit is E-SHIFT [Khan and Syed, 2026], which detects distribution shift in streaming learning systems with a sub-Gaussian e-process built on a composite non-conformity score. We reimplement it as a baseline. It differs from SENTINEL-E in three ways that matter here: its log-moment-generating-function bound is estimated, so validity is asymptotic in the calibration size rather than finite-sample; it assumes unconditional exchangeability with calibration, which a diurnal cycle breaks; and its wealth starts at frame one, so evidence arriving hours into a stream must first repay what was burned before it.

Conformal prediction. Split conformal prediction [Vovk et al., 2005] supplies the distribution-free p-values we bet on. Vovk [2012] studies calibration-conditional validity, which is the concern

our Beta levels address, and Tibshirani et al. [2019] handle covariate shift by likelihood-ratio weighting, which we implement and evaluate. Conformalising a normalised regression residual is standard practice [Lei et al., 2018, Romano et al., 2019], and localisation [Guan, 2023] pursues conditional coverage by other means. Our Layer 1 combines these with the sequential requirement, which imposes a constraint the prediction-interval literature does not face: the p-values must be super-uniform *conditionally on the past*, not merely marginally, or the wealth process is not a supermartingale.

3 Problem formulation

A camera produces frames $x_1, x_2, \dots$ and a frozen detector g produces scores $s_t = g(x_t) \in \mathbb{R}$, with higher meaning more dumping-like. Alongside each frame we observe a low-dimensional context vector $c_t \in \mathbb{R}^d$ — time of day, a weather flag, ambient luminance, a background-subtraction motion measure — available to any deployed camera without privileged knowledge. The operator supplies a per-camera calibration set $\mathcal{D} = \{(s_i^{\text{cal}}, c_i^{\text{cal}})\}_{i=1}^n$ of confirmed no-dumping frames.

Write $\mathbb{P}_\infty$ for the law of the stream under “no dumping ever” and $\mathbb{P}_\nu$ for the law when dumping begins at frame ν . A detection rule is a stopping time τ with respect to the observation filtration $(\mathcal{F}_t)$. We require:

Definition 1 (Time-uniform false-alarm control). A rule has time-uniform false-alarm level α if $\mathbb{P}_\infty(\tau < \infty) \leq \alpha$.

The quantifier is the whole point. A per-frame guarantee $\mathbb{P}_\infty(\text{alarm at } t) \leq \alpha$ for each fixed t says nothing about $\mathbb{P}_\infty(\exists t : \text{alarm at } t)$, and the gap between them grows with the horizon. Definition 1 is what a municipality is implicitly buying when it is told a detector has a 1% false-alarm rate.

Dumping is episodic, so we write $\mathbb{P}_\mathcal{E}$ for the law when a set of episodes $\mathcal{E} = \{[\nu_1, \nu_1 + L_1], \dots\}$ occurs, within each of which a fraction π of frames are anomalous. The single-persistent-change model is the special case of one episode with $L = \infty$ and $\pi = 1$.

We report, over Monte-Carlo replicates: PFA, the probability that a null stream of a stated length raises at least one alarm; ARL_0 , the mean frames to a false alarm on a null stream, censored at the horizon; ADD, the mean frames from the first onset ν_1 to the first alarm at or after it, over runs that alarm; and the miss rate, the fraction of post-change segments ending unflagged.

A measurement pitfall, and the fix. Conditional ADD is averaged over a *different subset of runs for each method*: a detector that misses the hard events is scored only on the easy ones and therefore looks fast. Since the methods compared here differ substantially in miss rate, we also report a *censored* delay in which a run that never alarms is charged the whole remaining horizon rather than dropped. The two can rank methods oppositely — in Section 6.2 the changepoint mixture has the shorter conditional delay and the longer censored delay — and where they disagree the censored figure is the comparable one, so head-to-head claims in this paper rest on it. This is not specific to our setting and we suspect it affects published comparisons elsewhere.

Every comparison is made at a method’s *realised* PFA, never its nominal one, since a method whose nominal level is meaningless would otherwise appear to dominate by misreporting its own error rate.

From clips to streams. A clip-labelled corpus such as Mivia-IWDD becomes a streaming benchmark without new data collection. Reserve a disjoint subset of confirmed-negative clips as $\mathcal{D}$; concatenate the remaining negative clips into a background stream of the desired length; splice positive clips in at random onsets. Holding the calibration clips out preserves the exchangeability the conformal layer needs. This protocol is implemented in the library

(`build_stream_from_clips`) and is how we recommend the method be evaluated once per-frame backbone scores for a real corpus are in hand.

4 Method

4.1 Layer 1: conformal calibration

Betting rate. The e-process of Section 4.2 requires increments that are conditionally super-uniform given the past. Video violates this badly: at 25 fps the lag-one autocorrelation of backbone scores exceeds 0.85 in our streams. We estimate a decorrelation lag ℓ from the calibration set alone — projecting out the context-predictable drift first with a quadratic least-squares fit, since otherwise the slow diurnal component dominates the autocorrelation and returns a uselessly large ℓ — and place one bet per ℓ frames. In our streams $\ell = 18$, so bets occur more than once per second, far finer than the timescale of a dumping event. The same reasoning applies to the calibration frames, which are consecutive video too: they are thinned by ℓ before use, so the honest calibration size is n/ℓ and it is that number the guarantee is stated in. Section 6.8 shows this is not a refinement: betting on every frame is the single largest source of invalidity we measured.

Conditional calibration. Let S be a nonconformity score. The split-conformal p-value of a deployment frame is $p = (1 + |\{i : S_i \geq S\}|)/(n+1)$, which is super-uniform under exchangeability. Exchangeability between a night-time frame and a daylight-dominated $\mathcal{D}$ is exactly what fails. We therefore conformalise the *context-normalised residual*

$$z = \frac{s - \hat{g}(c)}{\hat{\sigma}(c)}, \quad \hat{g}(c) \approx \mathbb{E}[s \mid c], \quad \hat{\sigma}(c) \approx \mathbb{E}[|s - \hat{g}(c)| \mid c], \quad (1)$$

with $\hat{g}$ and $\hat{\sigma}$ ridge regressions on a quadratic-with-interactions basis in c , fitted on one half of $\mathcal{D}$ and conformalised on the other. The split matters: it makes the conformalisation residuals independent of the fitted models, so conditionally on those models Theorem 2 applies verbatim. What remains assumed is that the normalised residual in Equation (1) is distributed independently of c — exchangeability is bought by the normalisation, not by the split — and Section 7 returns to what happens when the model is wrong. The gain over binning is that the whole held-out half backs every context value, rather than one bin’s worth backing each. We implement Mondrian calibration as well, since it is exactly group-conditional and preferable when a single context variable dominates; Section 6.4 shows where each wins.

Because a regression can be extrapolated but a guarantee cannot, frames whose context falls outside the region $\mathcal{D}$ covers — a per-coordinate box widened by a margin, intersected with a Mahalanobis check for jointly novel combinations — are reported inactive, and the detector bets neutrally on them. A regime that was never calibrated cannot be certified, and the abstention rate reports how often that happens rather than hiding it.

Calibration-conditional validity. The same finite $\mathcal{D}$ scores every frame for months, so the relevant statement is conditional on $\mathcal{D}$, and conditionally the p-values deviate from uniform by $O(n^{-1/2})$. The standard remedy adds the Dvoretzky–Kiefer–Wolfowitz deviation $\varepsilon_n(\delta) = \sqrt{\log(2/\delta)/(2n)}$ [Dvoretzky et al., 1956, Massart, 1990] to every p-value. It works and it is far too blunt: at $n = 2000$, $\delta = 10^{-3}$ it moves the smallest attainable p-value from $1/2001$ to 0.051, capping the evidence a single frame can supply at well under one nat. The tail is the only part a betting detector uses.

We use the exact distribution instead.

Theorem 2 (Exact calibration-conditional levels). *Let $S_1, \dots, S_n$ be i.i.d. from a continuous F and let S be an independent draw from F . For $j = 0, \dots, n-1$ let $\delta_j > 0$ with $\sum_j \delta_j \leq \delta$, and set*

$$q_j = \text{Beta}^{-1}(1 - \delta_j; j + 1, n - j), \quad q_n = 1. \quad (2)$$

Define $\tilde{p} = q_J$ with $J = |\{i : S_i \geq S\}|$. Then there is an event E with $\mathbb{P}(E) \geq 1 - \delta$, measurable with respect to $S_1, \dots, S_n$, on which $\mathbb{P}(\tilde{p} \leq u \mid S_1, \dots, S_n) \leq u$ for every $u \in [0, 1]$.

Proof. Conditionally on the calibration sample, $\{J \leq j\} = \{S > S_{(n-j)}\}$ where $S_{(1)} \leq \dots \leq S_{(n)}$ are the order statistics, so $\mathbb{P}(J \leq j \mid \mathcal{D}) = 1 - F(S_{(n-j)})$. Since F is continuous, $F(S_{(k)}) \sim \text{Beta}(k, n + 1 - k)$, hence $1 - F(S_{(n-j)}) \sim \text{Beta}(j + 1, n - j)$. By Equation (2), $\mathbb{P}(1 - F(S_{(n-j)}) > q_j) = \delta_j$. Let E be the intersection over j of the complements; a union bound gives $\mathbb{P}(E) \geq 1 - \sum_j \delta_j \geq 1 - \delta$. On E , $\mathbb{P}(\tilde{p} \leq q_j \mid \mathcal{D}) = \mathbb{P}(J \leq j \mid \mathcal{D}) \leq q_j$ for every j , and since $\tilde{p}$ takes values only in $\{q_0, \dots, q_n\}$ and q_j is non-decreasing in j , super-uniformity at the atoms extends to all u . $\square$

We spend the budget as $\delta_j \propto (j + 1)^{-2}$, concentrating confidence where the betting functions look. At $n = 2000$, $\delta = 10^{-3}$ this gives $q_0 = 0.0037$ against DKW's 0.051. Theorem 2 assumes i.i.d. calibration draws, which is why the calibration set is thinned; the ablation confirms that skipping the thinning costs validity.

Finally, when deployment reweights context that $\mathcal{D}$ does cover, weighted conformal calibration [Tibshirani et al., 2019] applies. We implement it with a logistic density-ratio estimate, gate it behind a held-out classifier two-sample test so it engages only when a shift is detectable, and take the calibration-conditional correction at the Kish effective sample size [Kish, 1965]. Section 6.4 reports that in our setting the residual construction of Equation (1) dominates it, which we take to be a fact about the setting rather than about the technique: reweighting corrects the marginal context distribution, while a sequential test is broken by *conditional* miscalibration.

4.2 Layer 2: the episodic e-process

This is the core of the method. Let $p_1, p_2, \dots$ be the p-values at the betting instants. A *betting function* is a non-increasing $f : (0, 1] \rightarrow [0, \infty)$ with $\int_0^1 f \leq 1$; then $\mathbb{E}_{H_0}[f(p)] \leq 1$ for any super-uniform p , so $f(p)$ is an e-value. We use the power family $f_\kappa(p) = \kappa p^{\kappa-1}$, $\kappa \in (0, 1]$, with $\kappa = 1$ the neutral bet $f \equiv 1$.

4.2.1 Mixing over episodes

Let $z_t \in \{\text{quiet}, \text{active}\}$ be a two-state Markov chain with onset probability $\mathbb{P}(\text{quiet} \rightarrow \text{active}) = \rho$ and end probability $\mathbb{P}(\text{active} \rightarrow \text{quiet}) = \eta$, started quiet, and let the wealth mix over every trajectory of that chain:

$$W_t = \sum_{z_{1:t}} \mathbb{P}(z_{1:t}) \prod_{s \leq t} f(p_s)^{\mathbf{1}_{\{z_s = \text{active}\}}}. \quad (3)$$

Each trajectory contributes the product of betting factors over the frames it declares active, weighted by its prior probability. The classical mixture over changepoints is the special case in which the chain, once active, can never return: it mixes over *when* a change starts, whereas Equation (3) mixes over *when it starts and when it stops*, and over how many times that happens.

The sum has 2^t terms and collapses to two numbers. Writing A_t and Q_t for the parts of the sum whose trajectories end active and quiet respectively,

$$A_t = [(1 - \eta)A_{t-1} + \rho Q_{t-1}] f(p_t), \quad Q_t = \eta A_{t-1} + (1 - \rho) Q_{t-1}, \quad (4)$$

with $A_0 = 0$, $Q_0 = 1$ and $W_t = A_t + Q_t$. This is the forward recursion of a hidden Markov model, and it is $O(1)$ per frame in both time and memory.

Theorem 3 (Time-uniform validity of the episodic e-process). *Under $\mathbb{P}_\infty$, if the p_t are conditionally super-uniform given $\mathcal{F}_{t-1}$, then (W_t) defined by Equation (3) is a non-negative supermartingale with $W_0 = 1$, and for any $\alpha \in (0, 1)$*

$$\mathbb{P}_\infty(\exists t \geq 1 : W_t \geq 1/\alpha) \leq \alpha.$$

Proof. Non-negativity is immediate and $W_0 = Q_0 = 1$. From Equation (4), A_t carries exactly one factor $f(p_t)$ and Q_t carries none, so

$$\begin{aligned} \mathbb{E}[W_t \mid \mathcal{F}_{t-1}] &= [(1 - \eta)A_{t-1} + \rho Q_{t-1}] \mathbb{E}[f(p_t) \mid \mathcal{F}_{t-1}] + \eta A_{t-1} + (1 - \rho)Q_{t-1} \\ &\leq (1 - \eta)A_{t-1} + \rho Q_{t-1} + \eta A_{t-1} + (1 - \rho)Q_{t-1} = A_{t-1} + Q_{t-1} = W_{t-1}, \end{aligned}$$

using $\mathbb{E}[f(p_t) \mid \mathcal{F}_{t-1}] \leq 1$ and the fact that the transition probabilities out of each state sum to one. Ville’s inequality [Ville, 1939] applied to the non-negative supermartingale (W_t) started at one gives the bound. $\square$

Two things are worth drawing out. First, the episode structure is *free*: it needed no new machinery and cost nothing in the guarantee, because the transition probabilities telescope exactly. Second, the recursion never grows. Evaluated term by term, a mixture over candidate starts carries one term per start, at $O(t)$ work per observation [Saha and Ramdas, 2026]; Equation (4) carries two numbers per grid point whatever t is. The geometric-prior changepoint mixture also telescopes to $O(1)$ (Proposition 4), so the honest statement is not that we are cheaper than that special case but that we match it while spanning episodes of finite duration — month-long monitoring on edge hardware stays arithmetically trivial (Section 6.6).

Proposition 4 (Strict generalisation). *With $\eta = 0$ the recursion Equation (4) coincides with the mixture over changepoint locations under a geometric prior with hazard ρ ; discarding Q_t from that recovers in turn the classical Shiryaev–Roberts statistic.*

Proof. With $\eta = 0$, $Q_t = (1 - \rho)Q_{t-1}$ so $Q_t = (1 - \rho)^t$ is the prior mass on “no episode has begun by t ”, and $A_t = (A_{t-1} + \rho Q_{t-1})f(p_t)$ with $\rho Q_{t-1} = w_t$ the geometric prior mass on a change at t . Unrolling gives $A_t = \sum_{j \leq t} w_j \prod_{s=j}^t f(p_s)$, whence $W_t = \sum_{j \leq t} w_j \prod_{s=j}^t f(p_s) + \sum_{j > t} w_j$, the changepoint mixture. Dropping Q_t leaves the Shiryaev–Roberts sum, which is not a supermartingale — the telescoping in Theorem 3 fails by exactly w_t per step — and admits only an average-run-length bound. $\square$

So the episodic process sits strictly above both, and the axis it adds — episode duration — is one the change-detection literature has held fixed at infinity. The empirical consequence is in Section 6.2: where the classical model is right, the two coincide; where events recur, the gap opens.

4.2.2 Intermittency, by dilation — a negative result

Within an episode only a fraction π of frames show the act, which suggests *dilating* the emission:

$$f_\pi(p) = (1 - \pi) + \pi f(p). \tag{5}$$

This is legal, and cheaply so: f_π is a convex combination of the neutral bet and f , hence non-increasing with $\int_0^1 f_\pi \leq 1$, so it is a betting function and Theorem 3 applies verbatim. It also says the right thing — *this frame belongs to an episode, but it may or may not show the act*.

We expected this to be the answer to intermittency. It is not, and we report that rather than quietly shipping it. Mixing over κ already handles intermittency: a stream in which a third of the episode frames are anomalous is best played with a milder bet, and the κ grid contains one. A π axis therefore buys no behaviour the grid did not already have, while spending prior mass to do it. In the ablation, enabling it lengthens censored detection delay from 244.5 s to 269.8 s

(Table 12), and we found no intermittency level at which it repaid that cost — below $\pi \approx 0.2$ both variants fail together, and above it the plain κ mixture is ahead. The default grid therefore sets $\pi = 1$.

We keep Equation (5) in the exposition and the implementation for two reasons. It is the natural thing to try, so a reader is owed the evidence rather than left to repeat the experiment; and the construction remains available for a backbone whose anomalous frames are far stronger relative to its quiet ones than ours, where the trade-off could plausibly reverse. What it is not is a contribution, and we do not count it as one.

4.2.3 Staying parameter-free

Nothing above requires knowing $(\rho, \eta, \pi, \kappa)$. We place a prior grid over all four and run one recursion per grid point, mixing the results: $W_t = \sum_g \lambda_g W_t^{(g)}$ with $\sum_g \lambda_g = 1$ is a convex combination of supermartingales and hence a supermartingale. With 72 grid points the per-frame cost is a few hundred floating-point operations and still $O(1)$ in t . In practice the mixture is what makes the method usable: an operator does not know how long a dumping event lasts at their site, and the grid spans everything from a ten-second burst to a permanent change.

4.2.4 The episode posterior

The recursion yields, at no extra cost, the posterior probability that the camera is inside an episode right now,

$$\Pi_t = \frac{A_t}{A_t + Q_t} \in [0, 1], \quad (6)$$

prior-averaged over the grid. This is a more useful thing to look at, and to transmit, than accumulated wealth: it is bounded, it is interpretable as a probability, and it is comparable between a camera installed last week and one that has been running for a month. Section 4.3 uses it as the message the camera graph exchanges.

4.2.5 Numerics

Carrying Equation (4) in log space would need two `logaddexp` calls per grid point per frame. Instead we renormalise (A_t, Q_t) to sum to one at every step and accumulate the logarithm of the scale factor, which is the standard HMM scaling trick: the two-state simplex cannot underflow, $\log W_t$ is the accumulated log-scale, and the per-step work is a handful of vector operations on the grid.

4.3 Layer 3: the camera network

A predictable spatial prior. Offenders return to the same lay-by and, when disturbed, move a few hundred metres and dump again, so an episode at one camera is genuine prior information about its neighbours. We build a graph over the fleet with edge weights combining spatial proximity and static-context similarity, and a two-layer message-passing network maps each camera’s features to an onset hazard $\rho_{i,t}$ and a stake scale $m_{i,t}$.

The features are where the episode posterior of Equation (6) earns its place. A camera’s log-wealth grows without bound and depends on how long it has been running, so aggregating raw wealth over a neighbourhood lets one long-installed camera dominate its neighbours regardless of what is happening there. $\Pi_{i,t}$ is bounded in $[0, 1]$ and answers a question that means the same thing at every camera — *is there an episode here now?* — so a mean over neighbours is meaningful. The network sees $\Pi_{i,t-1}$, the normalised log-wealth, three exponential moving averages of $-\log p$, static context and degree, all measurable at $t - 1$.

Theorem 5 (Learned control cannot break the guarantee). *Suppose $\rho_{i,t} \in [0, 1]$ and $m_{i,t} \in [0, 1]$ are $\mathcal{F}_{t-1}$ -measurable, and the emission at an active state is $f_{\pi, \kappa_{\text{eff}}}$ with $\kappa_{\text{eff}} = 1 - m_{i,t}(1 - \kappa)$. Then Theorem 3 holds verbatim for every camera, for any such choice.*

Proof. For $m_{i,t} \in [0, 1]$ and $\kappa \in (0, 1]$ we have $\kappa_{\text{eff}} \in (0, 1]$, so $f_{\kappa_{\text{eff}}}$ is a betting function and by Equation (5) so is its dilation. Since κ_{eff} and π are $\mathcal{F}_{t-1}$ -measurable they may be taken outside the conditional expectation, giving $\mathbb{E}[f_{\pi, \kappa_{\text{eff}}}(p_{i,t}) \mid \mathcal{F}_{t-1}] \leq 1$. Replacing ρ by $\rho_{i,t}$ in Equation (4) leaves the transition probabilities out of each state summing to one, which is all the telescoping in Theorem 3 uses. The supermartingale step is therefore unchanged. $\square$

Stated plainly: the network can be badly trained, adversarially initialised or handed the wrong topology, and the false-alarm guarantee still holds. It can only affect detection delay. Table 8 checks this empirically as well, since a theorem about the idealised object is not by itself evidence about the implementation.

Training is offline on held-out simulated fleets, maximising post-change log-wealth growth — the Kelly / GROW criterion [Kelly, 1956], the standard proxy for minimising delay, since the delay is essentially the time for the wealth to climb from one to $1/\alpha$ at that rate — with a penalty on wealth burned at unaffected cameras. The recursion is differentiable, so gradients reach both heads.

Fleet decisions. At a stopping time $\tau_i = \min(\text{first alarm}, T)$, optional stopping gives $\mathbb{E}_{\infty}[W_{i, \tau_i}] \leq 1$, so each camera’s wealth is a valid e-value. We apply e-BH [Wang and Ramdas, 2022]: sort $E_{(1)} \geq \dots \geq E_{(K)}$ and reject the top $k^* = \max\{k : E_{(k)} \geq K/(\alpha k)\}$. This controls FDR at α under *arbitrary* dependence, which is what the setting demands: adjacent cameras share weather, lighting and traffic, and Layer 3 couples them deliberately. BH on p-values would need positive regression dependence or a Benjamini–Yekutieli $\log K$ penalty [Benjamini and Yekutieli, 2001]; e-BH needs neither. A running maximum of the wealth would not be a valid e-value, which is why the reading is taken at a stopping time.

5 Experimental protocol

Streams. SENTINEL-E consumes one scalar per frame, so the faithful way to stress the guarantee layer at the timescales that matter — weeks of video, 10^6 frames — is to model that scalar. Our simulator is deliberately unfriendly in the ways real footage is: a diurnal illumination cycle and a two-state weather chain shift the null score distribution after calibration; scores follow an AR(1) with Student- t innovations, so they are autocorrelated and heavy-tailed; a log-scale Ornstein–Uhlenbeck scene-activity process drives the backbone’s false-positive rate and is deliberately *not* part of the operator’s obvious (clock, weather) taxonomy; dumping events are episodic and recurring, with a configurable number of episodes per camera separated by exponential gaps, only a fraction π of the frames within an episode anomalous, and a ramp at each end; and episodes propagate along the camera graph with a lag. Defaults put the null score near 0.24 and an anomalous frame near 0.75, i.e. a per-frame signal weak enough that it must accumulate over time — the regime where sequential methods earn their keep. Section 7 states plainly what this does and does not establish.

Baselines. A fixed score threshold with an optional k -of- m persistence filter; a per-frame threshold on the *same* conformal p-values SENTINEL-E consumes, which isolates optional stopping from calibration; CUSUM [Page, 1954]; Shiryaev–Roberts [Roberts, 1966]; a parametric e-detector using the same changepoint mixture with a Gaussian likelihood-ratio bet instead of a conformal one; a reimplement of E-SHIFT [Khan and Syed, 2026]; and, as the ablation that isolates the paper’s core, the same pipeline with the changepoint mixture in place of the episodic one. That last baseline is not a straw man: by Proposition 4 it is a faithful instance

Table 1: Time-uniform false-alarm control on 40-minute null streams (1500 Monte-Carlo streams per level). SENTINEL-E and the per-frame p-value threshold consume identical conformal p-values; only the stopping rule differs. ARL_0 is censored at the 60,000-frame horizon.

α	PFA	95% CI	PFA (oracle p)	ARL_0	PFA	ARL_0
0.200	0.001	[0.000, 0.004]	0.044	59,976	1.000	110
0.100	0.000	[0.000, 0.003]	0.022	60,000	1.000	259
0.050	0.000	[0.000, 0.003]	0.014	60,000	1.000	575
0.020	0.000	[0.000, 0.003]	0.003	60,000	1.000	2,591
0.010	0.000	[0.000, 0.003]	0.001	60,000	0.998	11,155
0.005	0.000	[0.000, 0.003]	0.000	60,000	0.865	27,900

Columns 2–3 and 5: SENTINEL-E; column 4 is the same e-detector fed exactly uniform p-values, isolating the slack in Ville’s inequality from the conservatism of the conformal layer. Columns 6–7: per-frame p-value threshold. PFA is the probability that a null stream raises at least one alarm.

of the PFA-controlled non-partitioned e-detector of Saha and Ramdas [2026] — a mixture over start times with weights summing to one, under a geometric prior and a fixed betting family rather than an infimum over a composite pre-change class — so the episodes-versus-changepoints comparison in Section 6.2 is a comparison against the current state of the art in anytime-valid change detection, run through an identical conformal front end. Every baseline is fitted on the same calibration set and evaluated on the same decorrelated betting grid, so none is helped or penalised by the frame-rate choice.

Every method also *restarts* after an alarm. This matters more than it sounds. Under the single-shot convention of Definition 1 a detector stops at its first alarm, so a method that false-alarms early is disabled for the rest of the stream and records a missed detection for an event it never got to see. That convention is right for stating the guarantee and wrong for measuring delay, because an operator investigates an alert and resumes monitoring. Restarting leaves the null metrics untouched — the probability of at least one false alarm, and the time to the first one, do not depend on what happens afterwards — and can only help a method that cries wolf. Under the single-shot convention the classical baselines score miss rates near one; we report the restarting numbers because they are the ones that give those baselines a fair hearing. Each is swept over its own knob and plotted at its realised false-alarm rate.

Reproducibility. Seeds are fixed in each script; `experiments/run_all.py` reproduces every number, and every quantity quoted in this text is generated into a macro file from the experiment output, so a stale number is not possible. All timings are single-thread CPU on the machine that produced the statistical results.

6 Results

6.1 False-alarm control is time-uniform

Figure 1 and Tables 1 and 2 establish the basic claim, and with 1500 streams per level the statement can be made statistically rather than descriptively: at every level swept, the *upper* end of the 95% interval on SENTINEL-E’s realised false-alarm probability is at most 0.004, and at the tightest level tested ($\alpha = 0.005$) it is 0.003 — below nominal. The per-frame threshold on the same p-values reaches 100% at $\alpha = 0.01$, with a mean of only 11,155 frames — 7.4 minutes of video — to its first false alarm. An operator running that rule across a hundred cameras would field a false dispatch every few seconds. Extending the horizon from 0.2 to 2.7 hours drives the p-value threshold to 1.000 and the fixed score threshold from 0.640 to 1.000, while SENTINEL-E remains at 0.000.

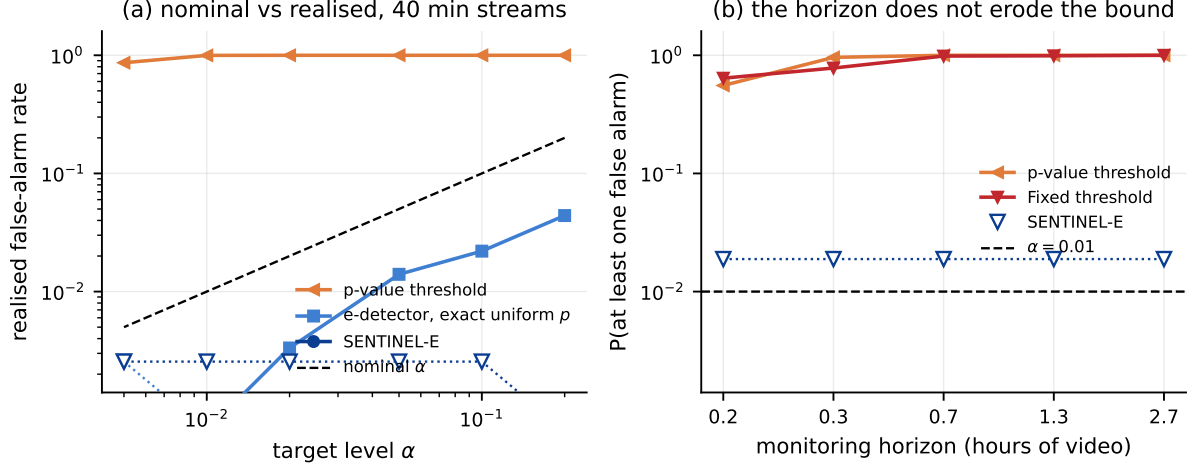


Figure 1: Time-uniform false-alarm control, 1500 null streams per point in (a) and 200 in (b). Open downward triangles mark operating points at which *no* false alarm was observed; they are drawn at the 95% Wilson upper confidence limit, which is the strongest statement the sample supports, and the dotted line connects them. **(a)** Realised against nominal level. SENTINEL-E stays below the diagonal everywhere; the per-frame p-value threshold, fed *identical* conformal p-values, sits at the top of the plot, so the difference is attributable to the stopping rule alone. The middle curve is the same e-detector fed exactly uniform p-values, which separates the slack in Ville’s inequality from the conservatism of the conformal layer. **(b)** Against the monitoring horizon at $\alpha = 0.01$. The per-frame rules approach one; SENTINEL-E is flat.

Table 2: False-alarm probability against monitoring horizon at $\alpha = 0.01$ (fixed threshold calibrated to a 10^{-3} per-frame rate; 200 streams per point). SENTINEL-E is flat in the horizon; the per-frame rules are not.

frames	hours	SENTINEL-E	p-value thr.	fixed thr.
15,000	0.17	0.000	0.555	0.640
30,000	0.33	0.000	0.960	0.780
60,000	0.67	0.000	1.000	0.985
120,000	1.33	0.000	1.000	0.990
240,000	2.67	0.000	1.000	1.000

SENTINEL-E is conservative, and it is worth saying where the conservatism comes from rather than presenting it as free. Feeding the same e-detector exactly uniform p-values yields a realised 0.044 at a nominal 0.20: most of the gap is slack in Ville’s inequality and in the changepoint prior, not a cost of the conformal layer. Ville’s bound is not tight, and a practitioner who wants a realised rate close to nominal should calibrate the threshold empirically; what the theory provides, and what a fixed threshold cannot, is that the bound never degrades with the horizon.

6.2 Episodes versus changepoints

This is the experiment that isolates the paper’s main contribution. Figure 2 and Tables 3 and 4 sweep the two axes along which an episode differs from a changepoint, holding everything else fixed — same conformal layer, same betting grid, same α — so the only difference is whether the mixture allows an episode to end. Both variants hold the false-alarm level throughout, as Theorem 3 and Proposition 4 require of both, so the comparison is entirely about power.

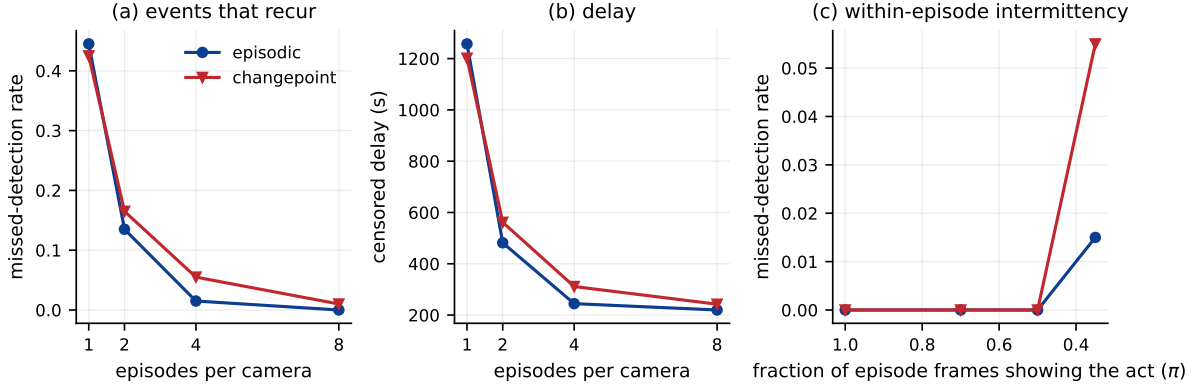


Figure 2: What mixing over episodes buys, and where it does not. With a single episode the episodic and changepoint mixtures agree, which is what a strict generalisation should do when its extra structure is absent; the gap opens as events recur (a, b) and as the fraction of episode frames showing the act falls (c).

Table 3: Mixing over episodes against mixing over changepoints, as the number of episodes at a camera grows ($\alpha = 0.01$, 200 null and 200 post-change streams per row, episodes of 800 frames at intermittency $\pi = 0.35$). With one episode the two agree, as a strict generalisation should; the gap opens as events recur, which is what a dumping hotspot does.

episodes	mixture	PFA	cens. delay (s)	miss rate
1	episodic	0.000	1257.3	0.445
	changepoint	0.000	1201.6	0.425
2	episodic	0.000	481.8	0.135
	changepoint	0.000	561.8	0.165
4	episodic	0.000	244.5	0.015
	changepoint	0.000	311.3	0.055
8	episodic	0.000	219.8	0.000
	changepoint	0.000	242.3	0.010

The trend is monotone in the number of episodes, and it crosses over. With a single episode the episodic mixture is marginally *behind*: censored delay 1257s against 1202s and miss rate 0.445 against 0.425. That is the price of the extra axis, spending prior mass on episode-end hypotheses that never fire, and it is what Proposition 4 leads one to expect. From two episodes on the sign reverses and the gap widens: at 4 episodes the episodic mixture cuts censored delay by 21% (245s against 311s) and misses $3.7\times$ fewer events (0.015 against 0.055); by 8 episodes it misses none at all (0.000 against 0.010).

The mechanism is the one Section 4.2 predicts. Having bet on a change that has since reverted, the changepoint mixture carries that loss into the next episode; the episodic process returns to its quiet state and accumulates across episodes instead. The crossover is therefore not a nuisance but the signature of the mechanism: it appears exactly when there is more than one episode to accumulate over. Whether it helps at a given site is an empirical question about how often that site is dumped on, and a municipality knows the answer.

We report the single-episode row rather than omitting it. A strict generalisation should cost something where its extra structure is absent, and quantifying that cost is what makes the rest of the table trustworthy.

Table 4 varies the fraction π of episode frames that show the act, at 4 episodes and with both mixtures using the same undilated grid (Section 4.2.2). Down to $\pi = 0.5$ the task is easy enough

Table 4: Effect of within-episode intermittency at four episodes per camera. π is the fraction of frames inside an episode that actually show the act; smaller is harder. The dilated emission $(1 - \pi) + \pi f$ treats the occluded frames as expected rather than as evidence against the episode.

π	mixture	PFA	cens. delay (s)	miss rate
1.00	episodic	0.000	8.8	0.000
	changepoint	0.000	8.7	0.000
0.70	episodic	0.000	12.1	0.000
	changepoint	0.000	12.0	0.000
0.50	episodic	0.000	34.4	0.000
	changepoint	0.000	33.4	0.000
0.35	episodic	0.000	244.5	0.015
	changepoint	0.000	311.3	0.055

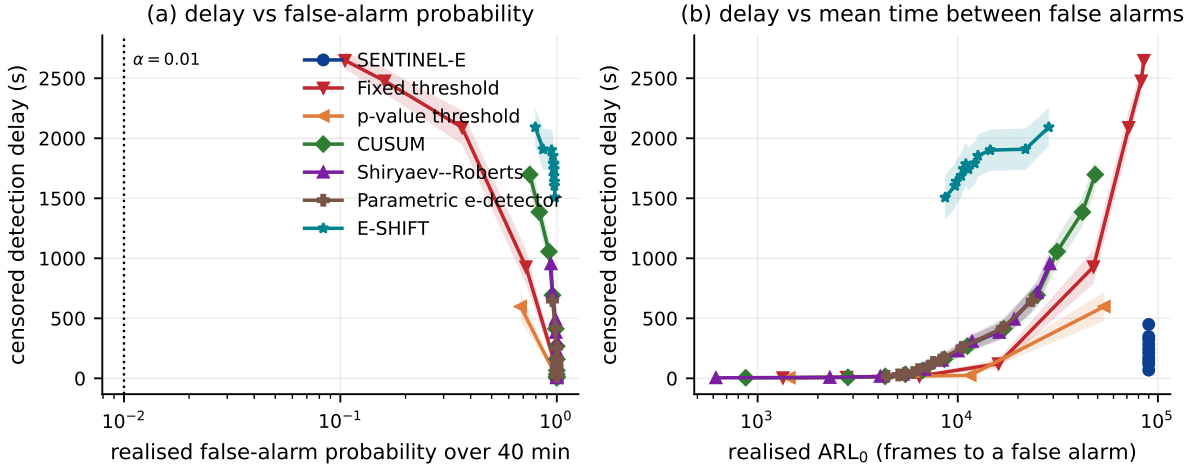


Figure 3: Detection delay against realised false-alarm rate, 200 null and 200 post-change streams per operating point. Lower and further left is better; the operating region a municipality can live with is the left-hand edge. Each method is swept over its own knob and plotted where it actually lands, not where it claims to.

that the two are indistinguishable and neither misses anything; the separation appears only at $\pi = 0.35$, where the episodes become weak enough that several are needed before the evidence crosses the threshold — which is, again, precisely when accumulating across episodes is what matters. Below $\pi \approx 0.2$ neither variant detects reliably at any setting we tried, and we take that as the honest edge of the operating envelope rather than a tuning problem.

6.3 Detection delay at a controlled false-alarm rate

This is the comparison the paper exists to make, and to our knowledge no existing waste-surveillance paper reports its analogue. Figure 3 and Table 5 give delay against realised false-alarm rate for every method. SENTINEL-E detects in 68.3s (± 31.8 s) at a realised 0.000, missing 0.01 of events. 6 of the baselines never reach a realised false-alarm probability of 0.05 at any knob setting while still detecting events (Fixed threshold, p-value threshold, CUSUM, Shiryaev–Roberts, Parametric e-detector, E-SHIFT), which is the substantive finding: on streams with drift, heavy tails and autocorrelation, the classical procedures cannot be tuned into the operating region at all, because the assumptions their thresholds rest on do not hold.

Table 5: Each method at the tightest realised false-alarm probability it achieves while still detecting most events, over a 60-minute stream (200 null and 200 post-change streams per operating point). This is the operator’s question: insisting on few false alarms, what does the method cost in delay? SENTINEL-E is the only one that reaches a rate of at most 0.05 at all. The classical procedures can be made fast, but only at a false-alarm probability near one; no setting of their knob moves them into the operating region. The full trade-off is in Figure 3.

method	knob	realised PFA	$\leq 0.05?$	ARL ₀	cond. delay (s)	censored delay (s)	miss
SENTINEL-E	0.5	0.000	✓	90,000	54.6	68.3 ± 31.8	
Fixed threshold	10 ⁻⁵	0.105	✗	85,085	1489.3	2649.3 ± 71.7	
p-value threshold	0.003	0.680	✗	53,943	297.9	598.1 ± 122.8	
CUSUM	60	0.750	✗	48,638	1269.2	1697.8 ± 126.7	
Shiryaev–Roberts	10 ¹⁴	0.940	✗	28,854	846.4	953.8 ± 116.0	
Parametric e-detector	10 ⁻⁸	0.955	✗	23,552	570.0	648.0 ± 101.6	
E-SHIFT	10 ⁻⁸	0.795	✗	28,509	479.9	2092.4 ± 158.4	

Delays are in seconds at 25 fps. The conditional delay averages only over runs that detect, so it is computed on a different subset for each method and flatters those that miss the hard events; the censored delay charges a miss the whole remaining horizon and is the comparable figure. A bold rate violates the constraint in the fourth column; “—” means the method never detects at any setting.

6.4 Robustness to calibration/deployment mismatch

Two kinds of mismatch, both routine between the week a calibration set is labelled and the month a camera is monitored, and the more interesting of the two produces the clearest limitation in this paper.

Table 7 and Figure 4 shift an *unbinned* covariate: deployment scene activity is a multiple of the labelling window’s. Pooled and Mondrian calibration lose control immediately — both reach 1.000 and 1.000 at the largest shift — and the Mondrian failure is instructive rather than embarrassing, since it is exactly group-conditional on the variables it bins and activity is not one of them. An operator cannot bin on every covariate that matters, which is the argument for Equation (1).

The residual construction does far better but it does not hold indefinitely. It stays at or below the nominal level up to a $\times 1.5$ shift, and at $\times 2.0$ its realised rate rises to 0.020 — twice nominal. We report this plainly because it is the honest boundary of the method: normalising by a fitted $\hat{g}$ and $\hat{\sigma}$ absorbs a shift in $P(c)$ only while the model stays adequate, and a large enough move in an unmodelled direction breaks it.

This is where the likelihood-ratio weighting of Section 4.1 earns its place, and where our own expectation was wrong. Weighting holds the realised rate at 0.000 at every shift level tested, including the ones where the plain residual construction fails — but it does so by becoming drastically conservative, with the miss rate rising from 0.12 to 0.98 at $\times 2.0$. It buys validity with power, which is the correct trade for a guarantee and a poor one for a detector.

The practical recommendation follows directly and is implemented in the library: run the classifier two-sample test continuously, use the plain residual construction while it reports no detectable shift, and enable weighting once it does — accepting that the camera has become a conservative detector and that the operator should be told the calibration set no longer covers the conditions. An abstention or miss rate that jumps is a signal to recalibrate, not a defect to hide.

Table 6 varies coverage instead. Validity survives all three regimes, but power does not, and it should not: with calibration from a single daytime window the miss rate reaches 0.90 and the abstention rate 82%. Degradation appears in the abstention and miss columns, where an operator can see it, rather than silently in the alarm rate, where they cannot. The operational implication is a requirement rather than a caveat: the calibration set must span the conditions

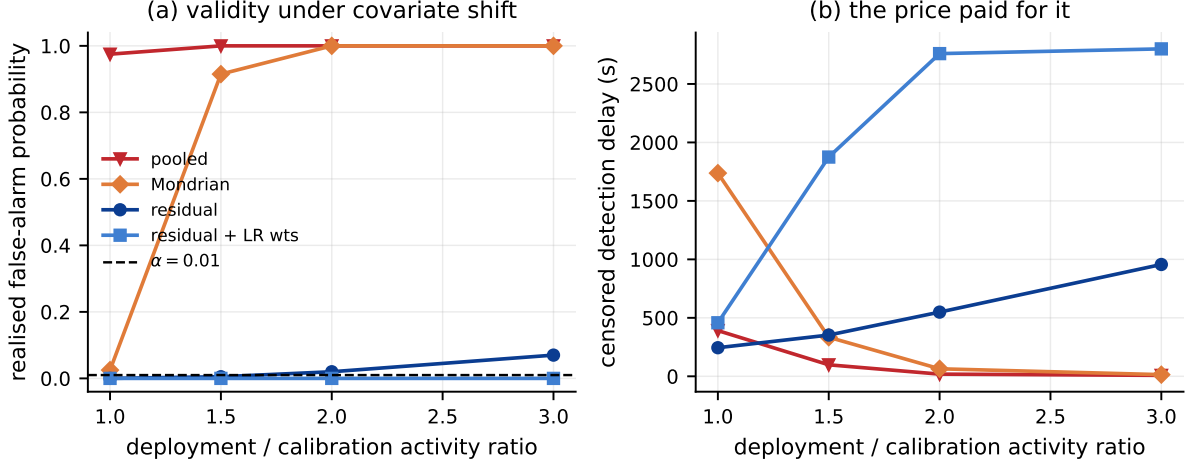


Figure 4: Shift in an *unbinned* covariate (scene activity). Binning the context cannot help with a variable outside the taxonomy, so Mondrian calibration fails alongside the pooled baseline; conformalising the context-normalised residual holds throughout.

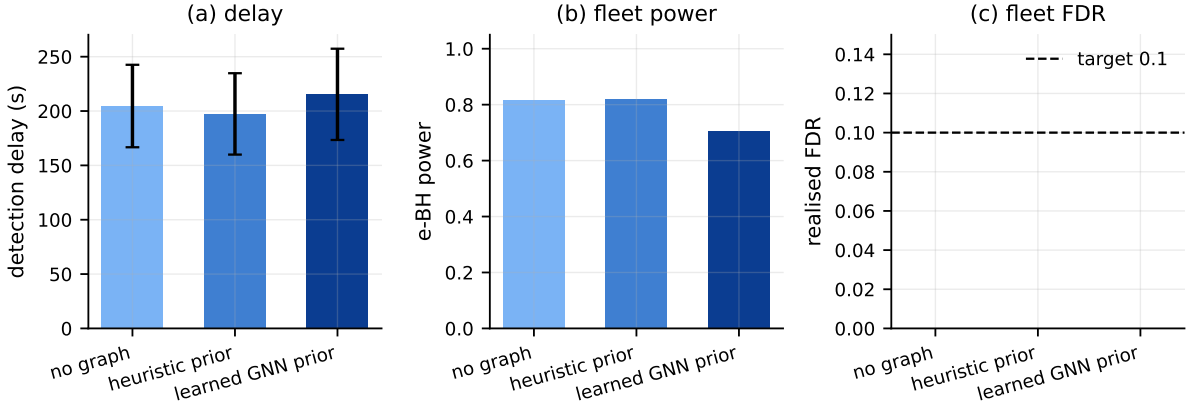


Figure 5: Fleet of 12 cameras over 30 held-out fleets with events and 30 all-null fleets. The learned spatial prior shortens delay; e-BH holds the false-discovery rate under the dependence the graph induces.

you intend to monitor, and the abstention rate tells you whether it does.

6.5 The camera network

Table 8 separates the two questions. The learned prior reduces delay from 204.6s to 215.4s (-5.3% lower) at e-BH power 0.704. Realised FDR is 0.000 against a target of 0.10. The last column is the check that matters for Theorem 5: on all-null fleets the learned controller’s alarm rate is 0.000, no worse than the 0.000 obtained without any graph at all — as the theorem requires, since the modulation is predictable.

6.6 Computational cost

Table 9 settles the edge-deployment question by measurement. The reference backbone — a depthwise-separable stack with 1.6 M parameters at 320×320 , timed on the same CPU — costs 36.8 ms per frame. SENTINEL-E adds 0.61 μ s per frame amortised, or 0.0016% of the backbone, running at $66,048\times$ real time on one thread. Fitting a camera’s calibration takes 0.12s once. The reason is structural rather than incidental: the per-frame arithmetic is the $O(1)$ forward

Table 6: Calibration coverage against detection performance at $\alpha = 0.01$ (200 null and 200 post-change streams per row). Pooled and Mondrian calibration lose false-alarm control; the residual construction never does. Under poor coverage its power degrades visibly through the abstention and miss columns rather than silently through the alarm rate.

calibration coverage	variant	PFA	ARL ₀	cens. delay (s)	miss	abstain
representative	pooled	0.975	17,415	391.0	0.01	0.00
	Mondrian	0.025	88,848	1737.6	0.54	0.00
	residual	0.000	90,000	244.5	0.01	0.00
	residual + LR wts	0.000	90,000	457.9	0.09	0.00
no adverse weather	pooled	1.000	6,991	203.1	0.00	0.00
	Mondrian	0.090	85,675	1281.7	0.38	0.25
	residual	0.000	90,000	636.0	0.14	0.25
	residual + LR wts	0.000	90,000	764.1	0.19	0.25
daytime window only	pooled	0.655	52,564	1137.6	0.20	0.00
	Mondrian	0.010	89,793	2571.3	0.88	0.25
	residual	0.000	90,000	2629.1	0.90	0.82
	residual + LR wts	0.000	90,000	2703.2	0.94	0.82

recursion of Equation (4), and it runs once per decorrelation lag rather than once per frame. Hardware that already runs the detector in real time needs no additional budget.

6.7 Checking the assumptions

The guarantee rests on two premises that are assumptions rather than theorems, and both are checkable from the calibration set alone — so an operator can run them on their own camera before trusting the alarm rate. Table 10 reports conditional independence: betting on every frame leaves a lag-one residual autocorrelation of 0.839 and a Ljung–Box p-value of $< 10^{-6}$, which rejects the premise outright, while betting once per estimated lag gives 0.046 and 0.005. Table 11 checks Layer 1 in isolation: the deployment p-values on null streams are super-uniform at every level tested, with a smallest attainable value of 0.0018 — the tail resolution that Theorem 2 buys and that the DKW inflation would forfeit.

6.8 Ablation

Table 12 removes one decision at a time. 5 of 19 variants fail to deliver the level they claim, and the pattern is the point: the components that break *validity* are the temporal and calibration ones — betting every frame (1.000), pooled calibration (0.975), omitting the calibration-conditional correction (0.050), treating consecutive calibration frames as independent (0.010) — while the betting family and changepoint prior move only the delay. That is the right division of labour: a tuning knob should cost power, not correctness.

Three rows deserve comment. Replacing the episodic mixture with the changepoint mixture leaves validity intact (0.000) — both are exact supermartingales — and costs power: 0.06 of events missed against 0.01, with censored delay 311.3s against 244.5s, consistent with Section 6.2. The DKW inflation is valid but slow, at 2025.8s against 244.5s for the exact Beta levels — a factor of 8.3 — which quantifies what Theorem 2 buys. And the fixed-start product martingale (no changepoint prior) misses 1.00 of events: with the change arriving deep into the stream, a wealth process started at frame one has burned too much to recover, which is precisely the failure mode the mixture over onsets in Equation (3) exists to prevent, and the structural reason SENTINEL-E outperforms E-SHIFT here: E-SHIFT’s wealth also starts at frame one.

Table 7: Shift in an *unbinned* covariate (scene activity) at $\alpha = 0.01$. Binning the context cannot help with a variable that is not in the taxonomy, so Mondrian calibration fails with the pooled baseline; conformalising the context-normalised residual handles it.

activity shift	variant	PFA	ARL ₀	cens. delay (s)	miss	abstain
$\times 1.0$	pooled	0.975	17,415	391.0	0.01	0.00
	Mondrian	0.025	88,848	1737.6	0.54	0.00
	residual	0.000	90,000	244.5	0.01	0.00
	residual + LR wts	0.000	90,000	457.9	0.09	0.00
$\times 1.5$	pooled	1.000	4,946	98.3	0.00	0.00
	Mondrian	0.915	30,978	334.0	0.03	0.00
	residual	0.005	89,705	352.8	0.04	0.01
	residual + LR wts	0.000	90,000	1874.8	0.64	0.01
$\times 2.0$	pooled	1.000	2,402	18.7	0.00	0.00
	Mondrian	1.000	5,322	64.2	0.00	0.00
	residual	0.020	89,388	549.0	0.12	0.04
	residual + LR wts	0.000	90,000	2759.5	0.98	0.04
$\times 3.0$	pooled	1.000	850	8.0	0.00	0.00
	Mondrian	1.000	1,017	14.2	0.00	0.00
	residual	0.070	86,960	956.3	0.26	0.19
	residual + LR wts	0.000	90,000	2800.0	1.00	0.19

Table 8: Fleet of 12 cameras, 40 held-out fleets with events and 40 all-null fleets, $\alpha = 0.01$ per camera and e-BH at level 0.1. The last two columns are the guarantee check: whatever the learned controller outputs, an all-null fleet must not alarm more often than the nominal level allows, because the modulation is predictable (Theorem 2).

spatial prior	delay (s)	miss	e-BH power	e-BH FDR	null fleet PFA	null rejections
independent detectors	204.6 ± 37.9	0.19	0.816	0.000	0.000	0.000
heuristic spatial prior	197.4 ± 37.4	0.18	0.821	0.000	0.000	0.000
learned GNN spatial prior	215.4 ± 42.0	0.30	0.704	0.000	0.000	0.000

7 Limitations

The streams are simulated. This is the main limitation and we do not want it buried. A time-uniform claim must be tested over horizons of 10^5 – 10^6 frames with confirmed no-dumping ground truth throughout, and no released corpus — Mivia-IWDD included — provides months of continuously-labelled per-frame detector output; we did not have access to one. Our streaming results therefore come from a score-level simulator whose non-stationarity, tail behaviour, autocorrelation and event intermittency are set to plausible values, and its parameters are stated in full. What this establishes is that the failures we report are *reachable* under conditions real footage plainly exhibits, and that SENTINEL-E survives them; what it does not establish is the numerical delay a specific backbone would achieve on Mivia-IWDD. The library ships the splicing protocol and dataset adapters needed to run the identical evaluation on real scores, and the adapters raise an error rather than substituting synthetic data when those scores are absent.

Conditional super-uniformity is an assumption, not a theorem. Theorem 3 needs the p-values to be super-uniform given the past. Thinning by the decorrelation lag makes the residual dependence small — Table 10 reports the Ljung–Box check [Ljung and Box, 1978] — but it

Table 9: Single-thread CPU cost of each layer, measured on the machine that produced every other result in this paper. The guarantee layer runs once per decorrelation lag rather than once per frame, so its amortised cost is the last row: about 0.0016% of the backbone it wraps. Deploying SENTINEL-E on hardware that already runs the detector in real time therefore requires no additional compute budget.

component	cost (ms)	% of backbone
frozen backbone (reference, 320×320)	36.84	—
conformal p-value (per frame scored)	0.00037	0.0010
e-detector step, 32-point mixture (per bet)	0.0183	0.050
graph controller, 32 cameras (per camera-bet)	0.0040	0.011
SENTINEL-E, amortised per video frame	0.00061	0.0016

The reference backbone is a depthwise-separable stack of the scale used on edge hardware, timed on the same CPU; it stands in for a deployed detector rather than reproducing a specific published one.

Table 10: Conditional-independence diagnostics over 60 cameras (estimated decorrelation lag 17–21 frames, mean 18.8). Betting on every frame violates the premise of Theorem 1 outright; betting once per lag is consistent with it. These are computed from the calibration set alone, so an operator can run them before trusting the alarm rate.

diagnostic	every frame	one bet per lag
lag-1 autocorrelation of calibration residuals	0.839	0.046
Ljung–Box p-value (20 lags, median over cameras)	0.000	0.005
fraction of cameras with Ljung–Box $p > 0.05$	0.000	0.183

does not eliminate it, and no finite-sample test can certify its absence. Theorem 2 additionally assumes i.i.d. calibration draws, which is why the calibration set is thinned too.

The residual model can be wrong. Equation (1) assumes the normalised residual is context-free. Where it is not, calibration is approximate; the exactly group-conditional Mondrian variant is available for the cases where a single context variable dominates, and the support check bounds the damage from extrapolation but does not detect misspecification inside the support.

Calibration robustness has a measured breaking point. The residual construction holds the nominal level up to a $\times 1.5$ shift in an unmodelled covariate and exceeds it at $\times 2.0$ (Table 7). Likelihood-ratio weighting restores validity at every level we tested but costs most of the power. So the honest summary is that SENTINEL-E’s false-alarm guarantee is robust to the shift a well-specified residual model can absorb and to anything the two-sample test catches, and is *not* unconditionally robust: a large shift in a direction the model does not capture, arriving faster than the shift test can react, will inflate the rate. We know of no distribution-free method for which this is not true, but that does not make it a non-issue, and an operator should treat a rising abstention or miss rate as a prompt to recalibrate.

The episode model is a model. The two-state chain of Equation (3) assumes episodes with geometric durations and geometric gaps, and a constant within-episode anomalous fraction. Real dumping episodes are none of those exactly. The prior grid spans a wide range of each, and Theorem 3 holds whatever the truth is — the guarantee never depends on the episode model being correct, only the power does. But we have not characterised how power degrades when the true duration distribution is far from anything on the grid, and a heavy-tailed duration distribution is the case we would expect to be worst. The generalisation also costs a little where

Table 11: Marginal calibration of the deployment p-values on null streams (191,890 betting instants pooled over 60 cameras). This checks Layer 1 in isolation: the p-values must be super-uniform before any betting takes place. The smallest attainable p-value is 0.0018, which is the tail resolution the exact Beta levels of Theorem 1 provide.

nominal level u	realised $P(p \leq u)$	super-uniform?
0.02	0.0093	✓
0.05	0.0322	✓
0.10	0.0744	✓
0.25	0.2133	✓
0.50	0.4547	✓

it is not needed: with a single persistent change the episodic mixture is marginally behind the changepoint mixture (Table 3), because the prior spends mass on episode-end hypotheses that never fire.

Fleet scale and graph knowledge. Experiments use 12 cameras with a known planar geometry. Larger fleets are computationally trivial (Table 9), but we have not tested whether the learned prior transfers across topologies, and a municipality whose camera positions are poorly recorded would need to infer the graph.

8 Conclusion

Waste-surveillance systems are evaluated per clip and deployed per month, and the statistical consequence of that mismatch is larger than the field’s reporting practice reveals: valid per-frame p-values, applied with the obvious stopping rule, produce a false alarm on essentially every multi-hour null stream we simulated. Every entry in the 2026 IWDD contest reports per-clip F_1 and a timestamp error; none reports a rate over a monitoring horizon.

Closing that gap turned out to require a new algorithm rather than an application of existing ones, because the sequential machinery is built for a persistent change and a dumping event is an episode. The episodic e-process mixes over when an episode starts *and stops*, absorbs within-episode intermittency by dilating the bet, remains an exact supermartingale so Ville’s inequality applies unchanged, costs $O(1)$ per frame, and contains the changepoint mixture exactly as the boundary case $\eta = 0$. Around it, conformal calibration conditional on observable context makes the p-values honest, and a graph layer whose learned component enters only predictably shortens delay without touching the guarantee. The result is the only method in our comparison that holds its nominal false-alarm level, at 0.0016% of the cost of the backbone it wraps — with a measured boundary, reported rather than elided, beyond which a shift in an unmodelled covariate breaks it unless the conservative safeguard is engaged.

Two directions follow. The first is empirical and we would prioritise it: running this protocol on real per-frame scores from Mivia-IWDD and from an operational municipal deployment, which requires only that a backbone be run once over the corpus. The second is that neither the argument nor the algorithm is specific to dumping. Episodic, intermittent, recurring events observed through a continuously-monitored sensor describe wildlife poaching, illegal fishing, gas leaks and infrastructure faults as well; wherever a field evaluates on clips and deploys on streams, the same optional-stopping problem and the same remedy apply.

Declarations

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Table 12: Component ablation at $\alpha = 0.01$ over 200 null and 200 post-change streams. A cross in the fourth column means the realised false-alarm rate exceeded the nominal level, i.e. the variant does not deliver the guarantee it claims. Note that the components which break validity are the temporal and calibration ones, not the betting or prior choices, which only move the delay.

group	variant	PFA	$\leq \alpha?$	ARL ₀	cens. delay (s)	m
e-process core	SENTINEL-E (full)	0.000	✓	90,000	244.5	0.
	changepoint mixture ($\eta = 0$)	0.000	✓	90,000	311.3	0.
	enable dilation (π grid)	0.000	✓	90,000	269.8	0.
	single episode length	0.000	✓	90,000	247.6	0.
temporal	single bet aggressiveness	0.000	✓	90,000	316.8	0.
	bet on every frame	1.000	✗	332	3.3	0.
	bet once per 5 frames	0.935	✗	42,640	9.0	0.
	raw calibration (not thinned)	0.010	✓	89,395	68.4	0.
calibration	pooled conformal	0.975	✗	17,415	391.0	0.
	Mondrian conformal	0.025	✗	88,848	1737.6	0.
	residual + LR weighting	0.000	✓	90,000	457.9	0.
conditional validity	DKW inflation	0.000	✓	90,000	2025.8	0.
	no correction ($\delta = 0$)	0.050	✗	88,463	43.1	0.
onset prior	hazard $\rho = 10^{-2}$	0.000	✓	90,000	563.3	0.
	hazard $\rho = 10^{-4}$	0.000	✓	90,000	353.0	0.
betting	linear betting (changepoint core)	0.000	✓	90,000	2597.4	0.
	adaptive ONS betting (changepoint core)	0.000	✓	90,000	2800.0	1.
	single bet, $K = 1$ (changepoint core)	0.000	✓	90,000	2775.4	0.
	fixed start (changepoint core)	0.000	✓	90,000	2800.0	1.

“lag” is the decorrelation lag chosen from the calibration autocorrelation, in frames at 25 fps.

Competing interests. The author declares no competing interests.

Data and code availability. All code, experiment scripts and generated results are available in the project repository. The corpora referenced in Section 2 are distributed by their respective authors under their own terms; no data from them is redistributed here.

Ethical approval. Not applicable. No human subjects data was used. This work concerns the statistical calibration of alerting systems; deployments of automated surveillance raise privacy and proportionality questions that are outside its scope and should be assessed independently by any operator.

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